

Ensemble-based Prototype-Augmented Multimodal Fusion for Ambivalence/Hesitancy Recognition

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Abstract

Ambivalence/hesitancy recognition in unconstrained videos is a challenging problem due to the subtle, multimodal, and context-dependent nature of this behavioral state. In this paper, a multimodal approach for video-level ambivalence/hesitancy recognition is presented for the 10th ABAW Competition. The proposed approach integrates four complementary modalities: scene, face, audio, and text. Scene dynamics are captured with a VideoMAE-based model, facial information is encoded through emotional frame-level embeddings aggregated by statistical pooling, acoustic representations are extracted with Emotion-Wav2Vec2.0 and processed by a Mamba-based temporal encoder, and linguistic cues are modeled using fine-tuned transformer-based text models. The resulting unimodal embeddings are further combined using multimodal fusion models, including prototype-augmented variants. Experiments on the BAH corpus demonstrate clear gains of multimodal fusion over all unimodal baselines. The best unimodal configuration achieved an average MF1 of 70.02%, whereas the best multimodal fusion model reached 83.25%. The highest final test performance, 71.42%, was obtained by an ensemble of five prototype-augmented fusion models. The obtained results highlight the importance of complementary multimodal cues and robust fusion strategies for ambivalence/hesitancy recognition. The source code will be publicly available after acceptance.

1. Introduction

Affective computing aims to endow intelligent systems with the ability to perceive, model, and interpret human affect from signals such as facial behavior [41], speech [13], language [7], and body motion [25]. This capability is important for human-computer interaction, digital health, education, and assistive technologies, where decisions often depend on subtle and context-dependent behavioral cues [33]. Within this area, the Ambivalence/Hesitancy

(A/H) Video Recognition Challenge of the 10th Workshop and Competition on Affective & Behavior Analysis in-the-Wild (ABAW) focuses on a particularly difficult binary task: given a video, the goal is to predict whether it contains A/H or not at the video level [14].

A/H recognition is important because these states are strongly linked to decision uncertainty, resistance, and fluctuating motivation during behavior change. In digital behavioral health interventions, such signals can help identify whether a person is ready to change, struggling with conflicting intentions, or at risk of disengagement [4]. Unlike basic emotions (such as happiness, surprise), A/H is subtle and often manifests through inconsistencies across modalities, for example, between what a person says, how they say it, and how they look while speaking. This makes the task inherently multimodal and particularly challenging [14, 33].

González-González et al. [14] established baselines using visual facial information, audio, and speech transcripts, and showed that text was the best unimodal cue, while multimodal fusion could yield additional gains but required specialized designs to capture cross-modal inconsistencies characteristic of A/H. Similarly, Savchenko and Savchenko [43] explored face, audio, and text modalities, with the best validation result achieved by combining textual and facial features, whereas Hallmen et al. [17] used text, vision, and audio, and their best final Behavioural Ambivalence/Hesitancy (BAH) submission was based on trimodal fusion. Therefore, multimodal modeling remains a promising direction for A/H recognition, especially when modality interactions are explicitly taken into account and fusion preserves modality-specific cues under uncertainty or partially inconsistent multimodal evidence [11, 52].

Beyond A/H-specific work, multimodal affective computing has highlighted the importance of modality-aware fusion. Prior work [16, 23, 24, 27, 28, 37–40, 53] suggests that integrating complementary visual, acoustic, and linguistic signals through dedicated fusion strategies can improve the robustness of affect-related prediction, while compact facial affect representations may serve as transferable features for downstream affective analysis.

This work proposes a multimodal approach that integrates audio, text, face, and scene information for video-level A/H recognition. First, a dedicated unimodal model is trained for each modality to learn compact modality-specific representations. The resulting modality embeddings are then projected into a shared latent space and fused by a Transformer-based multimodal module operating on modality tokens and augmented with a prototype-based classification objective. In this way, inter-modality dependencies are modeled directly at the video level, while the strengths of the specialized unimodal encoders are preserved for the final A/H prediction.

2. Related Work

González-González et al. [14] presented a broad benchmark suite including supervised unimodal, bimodal, and trimodal models based on facial video, audio, and speech transcripts, as well as zero-shot and personalization experiments. To study multimodal learning, they compared several fusion strategies, including concatenation-based fusion, co-attention, transformer-based fusion, and cross-attention fusion. Their results showed that text is the strongest unimodal cue, while effective A/H recognition requires specialized multimodal and temporal modeling to capture subtle inconsistencies across modalities. The official baseline reported for the current 10th ABAW challenge corresponds to a simpler zero-shot Multimodal Large Language Model (M-LLM) setup based on Video-LLaVA [29] with a simple prompt and vision modality only.

Among the approaches proposed for the first public A/H challenge, Hallmen et al. [17] used a trimodal architecture combining text, vision, and audio. Their framework extracts visual features with a Vision Transformer (ViT) [5], audio representations with Wav2Vec 2.0 [2], and transcript embeddings with the Bidirectional Encoder Representations from Transformers (BERT) [9] text encoder, applies temporal modeling to the visual and audio streams with Long Short-Term Memorys (LSTMs) [19], and fuses the resulting modality-specific representations with an Multilayer Perceptron (MLP) [36]. Their analysis showed that text is the most informative modality. The best final BAH submission was obtained using trimodal fusion, which is consistent with recent studies [11, 26] indicating that preserving complementary modality-specific information during fusion improves multimodal prediction.

In contrast, Savchenko and Savchenko [43] used a lighter pipeline based on facial, acoustic, and textual features extracted with EmotiEffLib [44], Wav2Vec 2.0 [2], and RoBERTa-based [30] text representations. Audio and text features were aligned to the visual timeline by interpolation, and fusion was implemented either through early concatenation followed by a feed-forward classifier or through blending of unimodal predictions. In addition, a video-level

Logistic Regression model over pooled text features was used as a filtering step. Their experiments showed that text was the strongest unimodal modality, while combining text and facial information yielded the best validation result.

The 10th ABAW submitted approaches mainly differ in the development of multimodal fusion strategies. Bekhouche et al. [3] used VideoMAE [47], HuBERT [20] and GoEmotions [6] for feature extraction from video, audio and text data, respectively. The authors also used modality-specific attention pooling and explicit pairwise conflict features. After that, they combined conflict-augmented early fusion with late text-guided fusion. Pereira et al. [32] also relied on SigLip2 [48] visual, HuBERT acoustic, and Foundation to Feature Large Language Model (F2LLM) [54] linguistic features, but added a fourth statistical modality and replaced a single fusion block with a committee of MLP, Random Forest, and Gradient Boosted Decision Trees (GBDT) classifiers over all modality combinations, combined by Particle Swarm Optimization (PSO)-based hard voting. Tang et al. [46] moved to the M-LLM setting: they split videos into short clips, fine-tuned Qwen3-Omni [51] on audiovisual prompt triplets, and aggregated clip-level predictions with max pooling and majority voting across multiple fine-tuned models.

In summary, prior work shows that text is the strongest unimodal cue for A/H, while multimodal fusion remains beneficial, especially when it preserves complementary information across modalities and accounts for uncertainty in multimodal evidence. Unlike previous approaches, which mainly rely on face, audio, and text with relatively simple fusion schemes, our approach additionally incorporates scene information and uses a Transformer-based fusion module over modality-specific representations learned by dedicated unimodal models, further regularized by a prototype-based classification objective.

3. Proposed Approach

The overall pipeline of the proposed approach is shown in Figure 1. Our approach addresses multimodal A/H recognition by extracting complementary information from several synchronized modalities. The obtained representations are then aggregated and fused to produce the final prediction. The following subsections describe the main components of the proposed approach.

3.1. Scene-based Visual Model

To analyze behavioral dynamics and detect uncertainty, we employ the Video Masked Autoencoder (VideoMAE) architecture [47] based on the ViT [10] framework and pre-trained on the Kinetics-400 corpus [21]. For each video, $T_v = 16$ frames are uniformly sampled, resized to 224×224 , and normalized using ImageNet [8] statistics. The input video clip is processed using tubelet embedding,

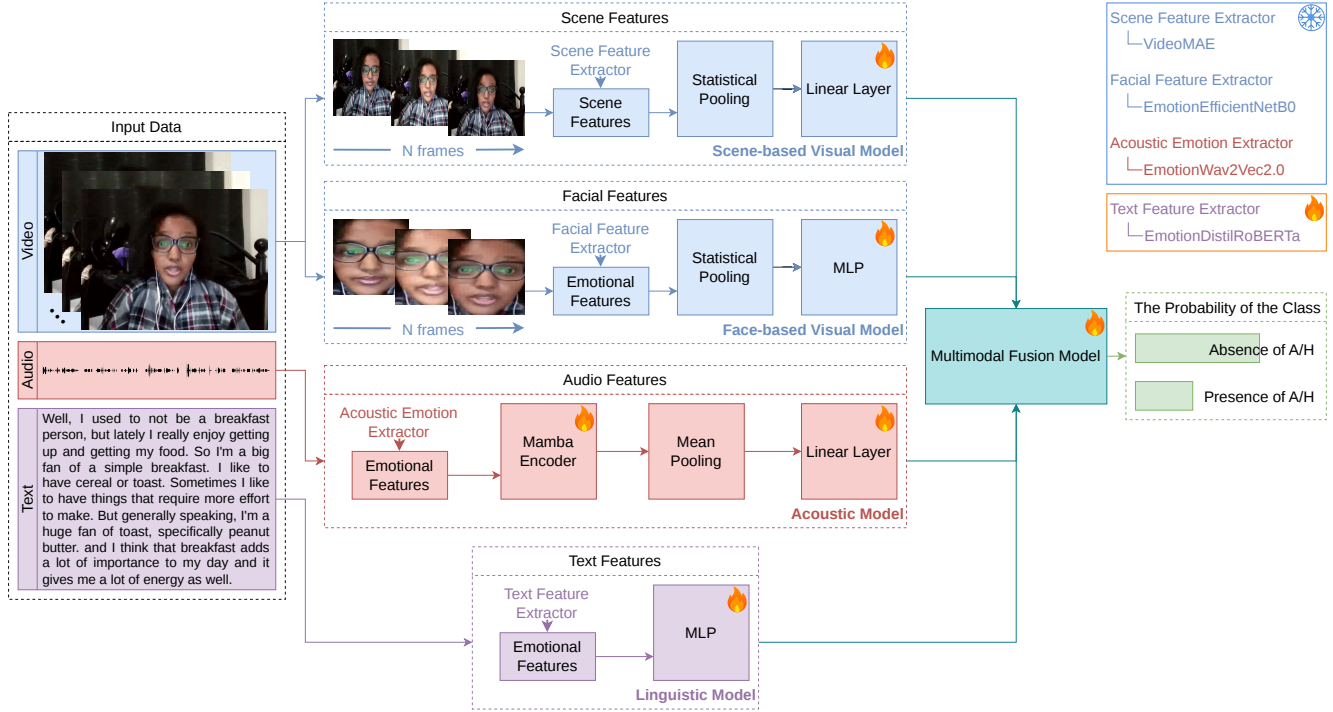


Figure 1. Pipeline of the proposed multimodal approach.

where it is partitioned into non-overlapping spatio-temporal patches of size $2 \times 16 \times 16$. These tubelets are projected into a $D = 768$ latent space and combined with learnable positional embeddings.

The resulting tokens are processed by a Transformer encoder with spatio-temporal self-attention to model spatio-temporal dependencies. A compact scene embedding h_s is obtained by applying global average pooling to the output tokens:

$$h_s = \frac{1}{N} \sum_{i=1}^N z_i, \quad (1)$$

where N is the number of output tokens and z_i denotes the representation of the i -th token.

3.2. Face-based Visual Model

For each video, frames are uniformly sampled, and a YOLO-based face detector¹ is applied to each sampled frame. When multiple faces are detected, the largest bounding box is selected. If no face is detected, the full frame is used as a fallback crop.

The cropped face is resized to 224×224 and normalized with ImageNet [8] statistics, then passed to an EfficientNetB0 [45] extractor fine-tuned on the AffectNet+ corpus [12], hereafter referred to as EmotionEfficientNetB0. The extractor produces one emotional embedding vector per

¹<https://github.com/lindevs/yolov8-face>

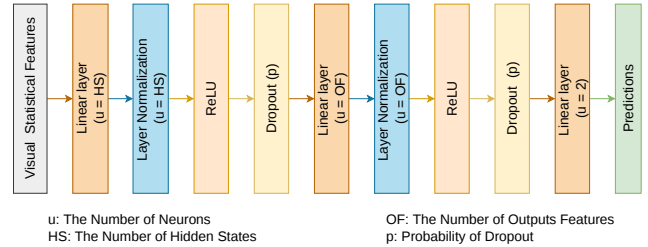


Figure 2. Visual MLP architecture.

sampled frame. No extra embedding normalization is applied.

For each video, frame-level embeddings $\{e_f\}_{f=1}^F$ are aggregated using statistical pooling:

$$\mu = \frac{1}{F} \sum_{f=1}^F e_f, \quad \sigma = \sqrt{\frac{1}{F} \sum_{f=1}^F (e_f - \mu)^2}. \quad (2)$$

The final face representation is formed by the concatenation $[\mu; \sigma]$. The resulting statistical representation is then used as input to an MLP, as shown in Figure 2.

3.3. Acoustic Model

For the audio modality, the audio track is extracted from each video and resampled to 16 kHz. A pre-trained

Wav2Vec2.0 model [50]², fine-tuned on the MSP-Podcast corpus [31], is used to extract acoustic emotion features. For simplicity, this encoder is referred to as Emotion-Wav2Vec2.0.

As a result, each audio segment is represented as a sequence of embeddings of size $T_a \times 1024$, where T_a denotes the number of temporal steps and 1024 is the feature dimension. The value of T_a depends on the duration of the input video.

The extracted acoustic representations are then processed by a sequential encoder to model temporal dependencies in the speech signal. In the main configuration, the extracted acoustic representations are processed by a Mamba encoder [15], followed by mean pooling over the temporal dimension to obtain a compact acoustic embedding:

$$a = \frac{1}{T_a} \sum_{t=1}^{T_a} s_t, \quad (3)$$

where s_t denotes the representation at the t -th temporal step and a is the pooled acoustic embedding. The resulting embedding is then passed through a linear layer to produce the final prediction. In addition to the Mamba-based [15] variant, a Transformer-based [49] encoder was also evaluated as an alternative sequential architecture.

3.4. Linguistic Model

For the linguistic modality, all audio recordings are transcribed into text. These transcriptions were automatically extracted by the corpus authors [14].

Several text-based modeling strategies are considered. A classical Term Frequency-Inverse Document Frequency (TF-IDF) [42] representation is first used to encode the relative importance of words and phrases in each transcription. The resulting features are combined with conventional classifiers, including Logistic Regression and Gradient Boosting Machines (GBM) models such as LightGBM [22] and CatBoost [34].

Pre-trained transformer-based language models are also used to obtain contextualized text representations. EmotionDistilRoBERTa³, allMiniLM⁴, EmotionTextClassifier⁵, and DeBERTa [18] are used to encode the transcriptions into dense embeddings. Both the [CLS] token representation and mean pooling over token embeddings are explored, followed by a GBM-based classifier.

²<https://huggingface.co/audeering/wav2vec2-large-robust-12-ft-emotion-msp-dim>

³<https://huggingface.co/j-hartmann/emotion-english-distilroberta-base/>

⁴<https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2>

⁵https://huggingface.co/michelleli99/emotion_text_classifier

In the main configuration, EmotionDistilRoBERTa is directly fine-tuned for A/H recognition. Its output representation is then fed into an MLP-based classification head to produce the final prediction. In addition, other fine-tuned transformer variants, including EmotionTextClassifier, are evaluated for comparison.

3.5. Modality Fusion Model

A two-stage strategy is adopted. First, unimodal models for scene, face, audio, and text are trained independently on the target corpus. Then, for each video, one fixed-dimensional embedding per modality is extracted from the corresponding unimodal branch and used as input to the fusion model.

Let M be the number of modalities, $m \in \{1, \dots, M\}$ the modality index, and $x_m \in \mathbb{R}^{d_m}$ the input embedding of modality m , where d_m denotes its original dimensionality. Each modality is projected into a shared latent space $\mathbb{R}^d$:

$$u_m = \phi_m(x_m), \quad u_m \in \mathbb{R}^d, \quad (4)$$

where ϕ_m is a modality-specific projector composed of a linear layer, layer normalization, Gaussian Error Linear Unit (GELU), and dropout. The resulting modality tokens are stacked into a matrix:

$$U = [u_1; \dots; u_M] \in \mathbb{R}^{M \times d}. \quad (5)$$

If one or more modality embeddings are unavailable for a given sample, a binary modality mask is provided to the fusion encoder so that the corresponding tokens are masked out during self-attention. A learnable modality embedding matrix $E_{\text{mod}} \in \mathbb{R}^{M \times d}$ is added to the token sequence:

$$Z^{(0)} = U + E_{\text{mod}}, \quad (6)$$

where $Z^{(0)}$ is the input token sequence to the Transformer encoder.

The token sequence is processed by a stack of Transformer encoder layers:

$$Z^{(l+1)} = T^{(l)}(Z^{(l)}), \quad l = 0, \dots, L-1, \quad (7)$$

where $T^{(l)}(\cdot)$ denotes the l -th Transformer encoder block and L is the number of layers. The final fused representation is obtained by masked mean pooling over the output modality tokens:

$$z_{\text{fused}} = \frac{\sum_{m=1}^M \mu_m z_m^{(L)}}{\sum_{m=1}^M \mu_m}, \quad (8)$$

where $z_m^{(L)}$ denotes the output representation of modality m at the last encoder layer and $\mu_m \in \{0, 1\}$ indicates whether modality m is available for the given sample.

For the prototype-augmented variant, the fused representation is compared with a set of learnable class-specific prototypes. Let $P_c = \{p_{c,k}\}_{k=1}^K$ be the prototype set for class

c , where K is the number of prototypes per class. The similarity score for class c is computed as:

$$\hat{y}_c^{\text{proto}} = \log \sum_{k=1}^K \exp \left(\frac{\tilde{z}_{\text{fused}}^\top \tilde{p}_{c,k}}{\tau} \right), \quad (9)$$

where $\tilde{z}_{\text{fused}}$ and $\tilde{p}_{c,k}$ denote the ℓ_2 -normalized fused representation and class prototypes, respectively, and τ is a temperature parameter. In the implemented model, the prototype head does not directly produce the final multimodal prediction; instead, it contributes an auxiliary loss term during training, while the final output logits are produced by the main linear classifier.

Accordingly, the overall training objective is defined as:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{proto}} \mathcal{L}_{\text{proto}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}}, \quad (10)$$

where $\mathcal{L}_{\text{cls}}$ is the main classification loss computed from the output logits of the linear classifier, $\mathcal{L}_{\text{proto}}$ is the auxiliary classification loss computed from the prototype-based logits, and $\mathcal{L}_{\text{div}}$ is the prototype diversity regularization term.

4. Experiments

4.1. Research Corpus

The BAH corpus is the core research corpus for the 10th ABAW A/H challenge. It was collected to support multimodal recognition of ambivalence and hesitancy in videos recorded in a realistic digital behavior change scenario [14]. Participants answered a predefined set of questions designed to elicit neutral, positive, negative, willing, resistant, ambivalent, and hesitant responses. The data were collected through an online platform with an avatar-guided interaction setup in order to mimic real-world personalized behavioral interventions [14].

The full corpus contains 1,427 videos from 300 participants, totaling 10.60 hours of recordings. It includes video-level and frame-level expert annotations, timestamps of A/H segments, cropped and aligned faces, speech transcripts with timestamps, and participant metadata [14]. Since ambivalence and hesitancy are difficult to separate reliably in practice, the task is formulated as a binary classification problem indicating the presence or absence of A/H [14]. For the 10th ABAW A/H challenge, the corpus is partitioned participant-wise into training, validation, public test, and private test subsets. Video-level prediction is evaluated using Macro F1-score (MF1) as the main metric.

4.2. Experimental Setup

During the development of the unimodal models, several alternative configurations were evaluated. Scene modeling was performed using 16-frame sequences resized to 224×224 , trained for 15 epochs with AdamW, a Learning Rate (LR) of $2e-5$, weight decay of $1e-2$, batch size 4,

cosine annealing, and Label Smoothing (LS) of 0.1. For the face-based visual modality, both statistical features with an MLP and raw embeddings extracted by ViT [10] and Contrastive Language-Image Pretraining (CLIP) [35] in combination with Transformer [49] and Mamba [15] encoders were considered. The best configuration was based on statistical features with an MLP. A grid search over the number of frames, hidden states, output features, LR, and optimizer selected a setup with 30 frames, 16 hidden states, 256 output features, a LR of $1e-3$, and AdamW.

For the acoustic modality, embeddings from different EmotionWav2Vec2.0 layers and different temporal encoders, including Mamba and Transformer, were evaluated. The best result was obtained with layer 10 and Mamba. In all primary runs, AdamW was used. The selected acoustic setup employed hidden size 256, feed-forward size 512, dropout 0.1, mean pooling, Mamba state size 8, convolution kernel size 4, and expansion factor 2. In the linguistic modality, TF-IDF features were evaluated with vocabulary sizes from 100 to 10,000 and n -grams from unigrams to trigrams, with hyperparameter tuning for Logistic Regression and GBM models. Transformer-based text models were also fine-tuned using partially frozen backbones and jointly trained MLP heads. The number of hidden layers varied from 1 to 3, the hidden size from 64 to 128, and dropout from 0 to 0.3. AdamW and Stochastic Gradient Descent (SGD) were explored, the LR was searched in the range from $1e-5$ to 0.1, the batch size was set to 16, and training lasted from 3 to 20 epochs with early stopping.

Multimodal fusion was performed using embeddings extracted from the selected scene, face, audio, and text models. Both non-prototype and prototype-augmented variants were evaluated, and the final system was based on the prototype-augmented fusion model operating on embedding-level inputs. To reduce sensitivity to random initialization, model selection relied on a stability-oriented hyperparameter search with Optuna [1] using five fixed random seeds: 42, 2025, 7777, 12345, and 31415. Each candidate configuration was trained and evaluated five times, and the final score was computed as the average MF1 across these runs. The selected fusion model used a shared latent dimensionality of 128, 6 Transformer encoder layers, 4 attention heads, a feed-forward expansion factor of 6, no [CLS] token, and dropout of 0.45. The prototype head used 16 learnable prototypes per class with temperature $\tau = 0.3$. Training was performed with RMSprop, a LR of $9.44e-5$, weight decay of $5.55e-4$, LS of 0.02, gradient clipping of 0.5, and a cosine learning-rate scheduler. The prototype loss weight was set to $\lambda_{\text{proto}} = 0.2$, while the diversity regularization term was disabled, i.e., $\lambda_{\text{div}} = 0$. Final predictions were obtained by averaging the class probabilities of the 5 seed-specific models.

ID	Model Configuration			BAH sub-corpus			
	Modality	Features	Classifier	Devel. (MF1, %)	Test (MF1, %)	Average (MF1, %)	Private test (MF1, %)
1	Face	EmotionEfficientNetB0 + Statistical Features	MLP	65.29	60.05	62.67	–
2	Scene	VideoMAE	Linear Layer	61.71	62.21	61.96	–
3	Audio	EmotionWav2Vec2.0 + Mamba	Linear Layer	67.20	70.87	69.03	–
4	Text	TF-IDF	Logistic Regression	68.30	67.75	68.03	–
5	Text	TF-IDF	CatBoost	65.56	72.02	68.79	–
6	Text	Fine-tuned EmotionTextClassifier	MLP	69.28	70.72	70.00	–
7	Text	Fine-tuned EmotionDistilRoBERTa	MLP	68.54	71.49	70.02	–
8	Models IDs 1, 2, 3 and 4	MFM	Linear Layer	80.79	77.03	78.91	–
9	Models IDs 1, 2, 3 and 5	MFM	Linear Layer	77.91	78.54	78.22	–
10	Models IDs 1, 2, 3 and 6	MFM	Linear Layer	78.35	77.03	77.69	–
11	Models IDs 1, 2, 3 and 7	MFM	Linear Layer	85.38	79.94	82.66	68.32
12	Models IDs 1, 2, 3 and 7	MFM with Prototype Head	Linear Layer	83.79	82.72	83.25	65.21
13	Models IDs 1, 2, 3 and 7	Ensemble of Five MFMs	Linear Layer	81.94	80.64	81.29	70.17
14	Models IDs 1, 2, 3 and 7	Ensemble of Five MFMs with Prototype Head	Linear Layer	83.00	80.77	81.89	71.42
Ablation Study							
15	Models IDs 1 and 2	MFM	Linear Layer	78.07	77.09	77.58	–
16	Models IDs 1 and 3	MFM	Linear Layer	63.36	71.44	67.40	–
17	Models IDs 1 and 7	MFM	Linear Layer	65.29	61.19	63.24	–
18	Models IDs 2 and 3	MFM	Linear Layer	77.37	77.66	77.51	–
19	Models IDs 2 and 7	MFM	Linear Layer	81.77	79.00	80.39	–
20	Models IDs 3 and 7	MFM	Linear Layer	67.05	70.99	69.02	–
21	Models IDs 1, 2 and 3	MFM	Linear Layer	76.10	79.15	77.62	–
22	Models IDs 1, 2 and 7	MFM	Linear Layer	79.89	77.65	78.77	–
23	Models IDs 1, 3 and 7	MFM	Linear Layer	68.08	70.41	69.25	–
24	Models IDs 2, 3 and 7	MFM	Linear Layer	79.89	77.63	78.76	–

Table 1. Experimental results on the BAH corpus for video-level A/H recognition. MFM refers to Multimodal Fusion Model.

4.3. Experimental Results

The experimental results are presented in Table 1. Among the unimodal models, the best average MF1 was obtained by the fine-tuned EmotionDistilRoBERTa model (70.02%), followed closely by the fine-tuned EmotionTextClassifier (70.00%) and the acoustic model based on EmotionWav2Vec2.0 and Mamba (69.03%). The TF-IDF-based text models also showed competitive results, reaching 68.03% with Logistic Regression and 68.79% with CatBoost. In contrast, the face- and scene-based models achieved lower average scores of 62.67% and 61.96%, respectively.

All multimodal fusion models outperformed the unimodal baselines. Among the single fusion models, the best average result was achieved by the prototype-augmented fusion model based on the selected scene, face, audio, and text modalities (Model ID 12), with 83.25%, while the corresponding fusion model without the prototype head (Model ID 11) reached 82.66%. These results indicate that both multimodal integration and prototype-based classification are beneficial under the development and public test settings.

The final test results show a different trend. Among the submitted four-modality systems, the best performance was obtained by the ensemble of five prototype-augmented fusion models (Model ID 14), which achieved 71.42%. The ensemble of five fusion models without the prototype head (Model ID 13) reached 70.17%, while the single fusion models achieved 68.32% (Model ID 11) and 65.21% (Model ID 12). Thus, although the prototype-augmented

single model achieved the highest average result, ensemble was essential for the strongest final test performance and improved robustness on the private evaluation split.

The ablation study of the best single model (Model ID 11) further confirms the benefits of combining modalities. The strongest two-modality result was obtained by combining scene and text features (Model ID 19), with an average MF1 of 80.39%. Among the three-modality settings, the best performance was achieved by two different models with combining modalities such as: (1) face, scene, and text (Model ID 22), with 78.77% and (2) scene, audio, and text (Model ID 24), with 78.76%. Overall, the best results were obtained with four-modality fusion.

Figure 3 presents two examples of the proposed multimodal model predictions (Model ID 11) together with the corresponding modality-wise attention patterns. The first example corresponds to the absence of A/H, where the model assigns a high confidence score of 98.12% and relies primarily on facial and textual features. The second example corresponds to the presence of A/H, where the model is also highly confident (99.66%) but shifts its attention toward audio and text modalities. These examples show that the contribution of each modality varies across samples and depends on the affective content of the input. Importantly, unlike unimodal models, the proposed multimodal model can adaptively combine complementary evidence from several modalities and reweight their importance for each sample, which is especially important for A/H recognition, where relevant cues may appear inconsistently across face, speech, language, and scene context.

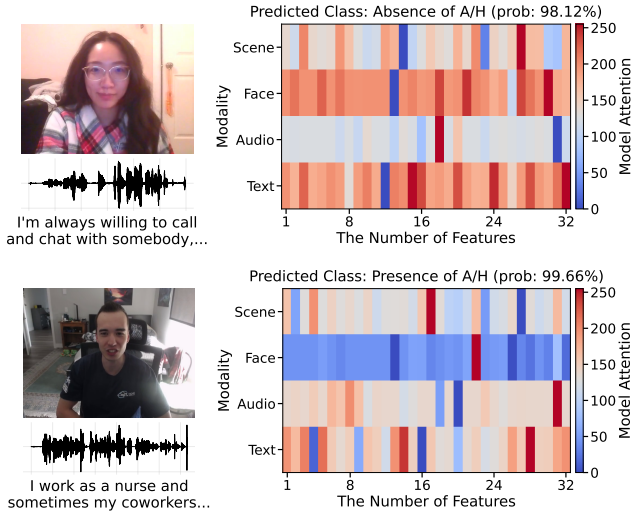


Figure 3. Visualization of the attention of the best single model (Model ID 11).

Table 2. Performance comparison (MF1) of the SOTA approaches on the private test subset. A, V, T and S are audio, video, text and scene modalities, respectively.

Approach	Modalities	MF1
Pereira et al. [32]	AVT	72.66
Bekhouché et al. [3]	AVT	71.51
Ours	AVST	71.42
Tang et al. [46]	M-LLM	67.48
Baseline	M-LLM	34.28

Table 2 demonstrates that the proposed approach achieves a competitive performance on the private test set, with an MF1 of 71.42%. This result is slightly lower than the best-performing approach by Pereira et al. [32], which achieved 72.66%, but it is close to the performance of Bekhouché et al. [3], this approach also achieved 71.51%.

5. Conclusions

This paper presented a multimodal approach for video-level A/H recognition on the BAH corpus. The proposed approach combined scene, face, audio, and text modalities and consistently outperformed unimodal baselines. Among the unimodal models, the best average MF1 of 70.02% was achieved by the fine-tuned EmotionDistilRoBERTa model, confirming the strong contribution of the linguistic modality. At the multimodal level, the best average result of 83.25% was obtained by the prototype-augmented fusion model, while the best final test performance of 71.43% was achieved by the ensemble of five prototype-augmented fusion models.

The ablation study showed that scene and text provide the strongest complementary signal among modality pairs, and that extending the fusion scheme to all four modalities yields the most effective overall solution. The comparison between single fusion models and their ensembles further indicates that robust model aggregation is important for generalization on the private test split. Overall, these results demonstrate that prototype-augmented multimodal fusion is an effective and robust strategy for A/H recognition in unconstrained videos.

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